DC-MHS: Datacenter-Secure Control Module (DC-SCM) 3.0 Specification Update

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Project Status

- DC-SCM 2.2 is in widespread use.
  - No further enhancements planned.

- DC-SCM 3.0 CLA group is operating
  - ~17 companies.
  - Targeting release initial ~0.5 package by EOY 2026; 1.0 by OCP GS 2027, containing:
    - Base spec (Doc + DC-SCI 3.0 + MCAD).
    - DC-SCM3.0-based HPM Common Circuit Design Spec Type 1 (single node)
    - TBD if LTPI base specification will be revised or not.
  - 1st product intercepts possible starting CY 2028.
  - Form factor + pinout are not Forward/Backward Compatible
    - Old-to-new and new-to-old mis-plug prevention.

DC-SCM 3.0 Goals

- Maximize interoperability for module re-use with less SCM-HPM co-design dependencies
- Introduce the Direct vs Indirect (bridged) host management concept
- Modular / Scalable Connectorization
  - Section 0 / Mini-SCI - BMC management card (dense designs, interposer typical)
  - Section 1 / Typical - Enterprise
  - Section 2 / Maximum - Either
    - Innovation Zone - Embedded I/O or more direct BMC-to-platform interfaces.
    - 2nd direct node - Q: Is 1 DC-SCM managing 2 bootable hosts of real interest?
  - Exact connector is TBD; Trending mini-SCI = M.2 & Typical=SFF-TA-1002 w/ offset shift; less than 4C+.
  - HPM FRU access = I2C or USB; Add-certificate (via M-PnP), New DMTF FRU spec alignments.

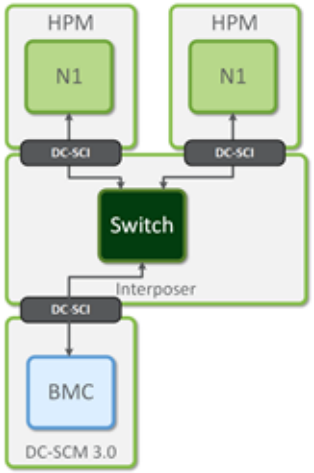
Maintain close alignment to:

- DC-MHS - All specs including + Modular-Plug-n-Play.
- OCP OBMF protocol and Interface Consolidation Protocol (ICP) specifications.
- Any potential HPM form factor effects.

Top Configs Considered

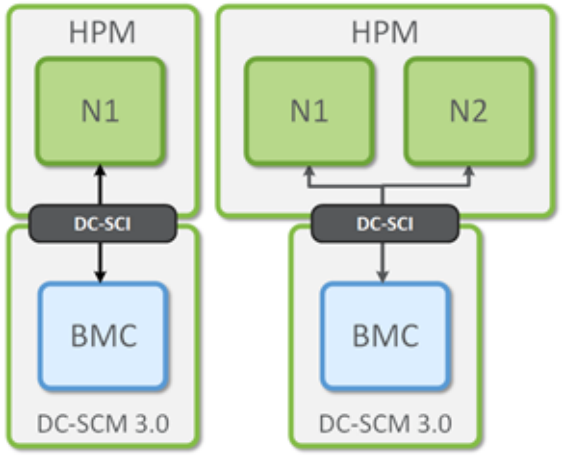
With Interposer

Conn Sec 0 +1 required, 2 optional



Direct Configuration

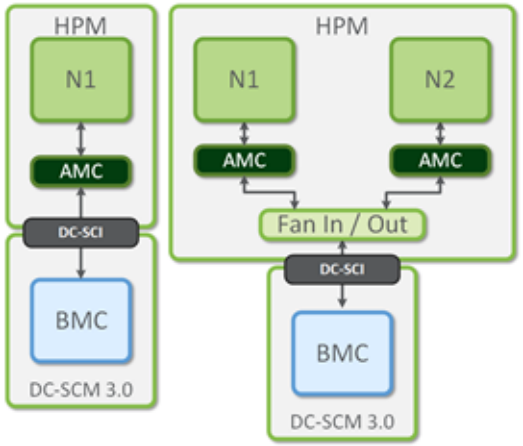
Conn Sec 0 & 1 required, 2 optional



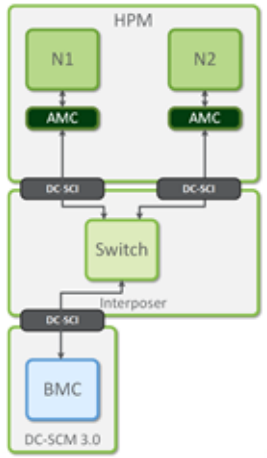
Real Demand?

Indirect Configuration

Conn Sec 0 only



Conn Sec 0 only



DC-SCI 3.0 (Secure Control Interface)

- Targeting as much single function/pin as possible for optimal HW interoperability with limited dual functions
- Trend toward USB bridging to HPM + peripheral interfaces. Aligned to Open Boot & Mgmt Framework (OBMF) enablement
- On-demand remote bus generation; Ideal bus bandwidth, Config scaling & best signal integrity for large systems
- Reduced PCIe bus lanes (X4X1X1->X1 per node)due to continued SOC integration (e.g., USB Host controller)
- Re-evaluation of optimal functional partitioning for ideal re-use, maximum density, cost (no default need for SiP/SoM)
- **Real-time & 100% available critical bit interfaces**:
  1) SGPIO - minimum real-time needs, minimal logic, low pin count, CRC.
  2) LTPI - WG seeks community feedback on functional requirements:
     (1) 1.0 Keep as-is protocol . Backward compatible logically speaking. No Link Layer.
     (2) 2.0 Enhanced support (Add Link layer + Random Access). Not backward compatible.
     (3) Deprecate

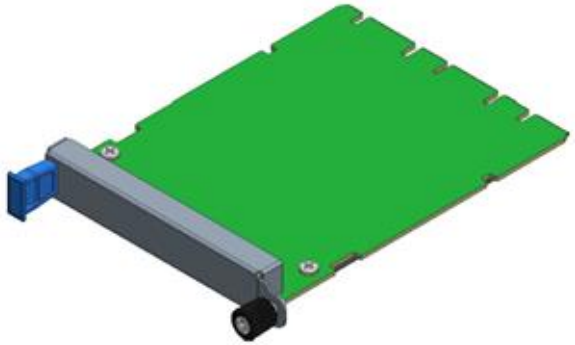
Interface Deltas (Tentative - not comprehensive)

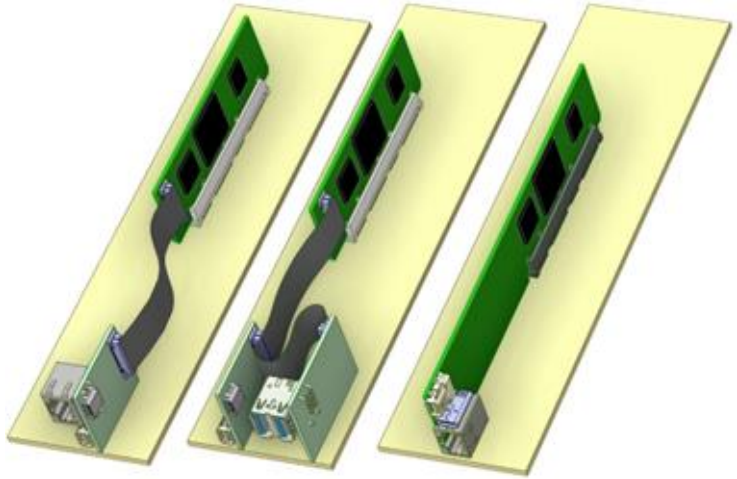
Bus Type	SCM2.2 (Single Node)	SCI 3.0 Section 0 (mini-SCI)	SCI 3.0 Section 1 (Typical/ additive)	Rationale
PCIe (Host domain)	X1, X1, X4	X1	0	BMC's integrate USB Host controllers or add PCIe switch. Almost no known adopters of X4
NC-SI / RMII	1X	0	0	Generate on demand
I3C	6X	0	0	USB-bridge improves bus scaling, generate on demand, best signal integrity
I2C	18X	0	TBD (1<=X<=few)	
Pwr	4X	2X	2X	Power scales for typical
USB	4X 2.0, 1X3.0	4X 2.0		Working through domains & directions
SGMII	1	1	0	
SGPIO / LTPI 1.0	1 / 1	0 / 0	1 /1	TBD on LTPI 2.0 deltas (+random access)
TPM SPI	2 types	0	1	Dedicated TPM SPI
eSPI	1	0	0 or 1	Foundational - Considering eSPI OR QSPI, not both, since eSPI functions unrelated to boot can be accomplished on HPM (FPGA/SMC).
QSPI	1	0	0 or 1	Mini-SCI → ROMs on HPM +
BMC_RECOVERY	0	1	0	New requirements

Mechanical Form Factors

- Keep popular Horizontal Form Factor (HFF) envelope
- Expect wider adoption of DC-SCM 2.2 introduced Compact Form Factor (CFF) with embedded or disaggregated I/O

- Evaluating an M.2 variant for the mini-SCI version. TBD on keying for old-new & new-old misplug prevention
- No further work planned on 2U vertical variants
- Evaluating Direct Liquid Cooling guidance
- Evaluating Common I/O possibilities (e.g., sunset VGA, mini-DP vs USB Type-C alt mode)





FEEDBACK NEEDED

Q1) Multi-node interest 1 BMC managing 2 host nodes?

Q2) Form factor insights

Q3) Long term sideband evolution views

Q5) Real-time & remote peripheral interface

USB only, USB+SGPIO, UBS+LTPI1.0, USB+LTPI2.0

Q6) Choose an Acronym for Indirect mode devices:

- AMC - Aux Management Controller
- SMC - Satellite Management Controller
- SMA - System Management Agent
- BIC - Bridge IC
- OBMF Bridge
- Other

Call to Action

- Join / Learn
  - To add https://ocp-all.groups.io/g/OCP-CLA-DC-SCM-3/
  - Server/MHS/DC-SCM-Specs-and-Designs - OpenCompute

Thank You!

